

Performance Results for PID Buck Converter

Performance at Noise Level $\sigma = 0.0$

Target Voltage (V)	Avg. Stabilisation Time (ms)	Avg. Overshoot (V)	Avg. Steady-State Error (V)
30.00	34.88 ± 0.00	24.903 ± 0.000	0.0978 ± 0.0000

Performance at Noise Level $\sigma = 0.001$

Target Voltage (V)	Avg. Stabilisation Time (ms)	Avg. Overshoot (V)	Avg. Steady-State Error (V)
30.00	34.84 ± 0.05	24.903 ± 0.000	0.0983 ± 0.0002

Performance at Noise Level $\sigma = 0.01$

Target Voltage (V)	Avg. Stabilisation Time (ms)	Avg. Overshoot (V)	Avg. Steady-State Error (V)
30.00	35.24 ± 0.39	24.775 ± 0.180	0.0964 ± 0.0023

Performance at Noise Level $\sigma = 0.1$

Target Voltage (V)	Avg. Stabilisation Time (ms)	Avg. Overshoot (V)	Avg. Steady-State Error (V)
30.00	36.88 ± 0.96	24.901 ± 0.310	0.1031 ± 0.0037

Analysis and Discussion of Results

The PID-controlled buck converter demonstrates a consistent and highly robust performance across all tested noise levels. The controller's response is characterized by its rapid speed and a critically high voltage overshoot.

Controller Response Characteristics

The dominant feature of the controller's performance is a very large **voltage overshoot**.

- In the noise-free case, the controller overshoots the 30V target by **24.9V**, causing the output to spike to nearly 55V. This behavior is indicative of a highly **underdamped system**, where the controller applies an aggressive, high-gain action to reach the setpoint as quickly as possible. In the context of a power converter, delivering almost double the intended voltage, even transiently, is a significant event.
- This aggressive strategy results in a very fast **stabilisation time** of approximately 35 ms, which remains consistent across the different noise levels.

Impact of Sensor Noise

A key finding is the controller's remarkable **robustness to sensor noise**.

- Unlike other control systems that can become unstable, this controller **successfully stabilizes** in every run, even at the highest noise level of $\sigma = 0.1$.
- The system exhibits a **graceful degradation** of performance as noise increases. The primary effect of noise is not a loss of stability but an increase in the **variability** of the results. This is visible in the growing standard deviation for all metrics, particularly the stabilisation time and steady-state error.
- The **steady-state error** remains relatively low and stable at around 0.1V across all test cases, showing that the controller effectively rejects noise and maintains its target precision once it has settled.

Conclusion

The results show a controller that prioritizes **speed and stability** over a smooth, non-overshooting response. It successfully regulates the voltage under significant sensor noise, albeit with a less consistent settling time as noise increases. The defining characteristic of its performance is the trade-off between its fast stabilisation time and the large initial voltage overshoot required to achieve that speed.